Cycles, Not Trends

A Time Series Analysis of Canadian Lynx Population Dynamics

Samundra Adhikari

Applied Time Series Project

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Abstract

This paper applies the full Box–Jenkins time series pipeline to the canonical `lynx` dataset, which records annual Canadian lynx trappings from 1821 to 1934. I document the absence of any long-run trend, demonstrate that classical Holt–Winters smoothing is a poor fit for a cyclical process, and derive valid trend inference via a Generalized Least Squares (GLS) correction for autocorrelated errors. Model identification via ACF/PACF analysis and AIC comparison across AR, MA, and ARMA specifications identifies ARMA(2,1) on the first-differenced log series—equivalently ARIMA(2,1,1) on log-transformed counts—as the preferred model, confirmed by a new ARIMA(2,1,0) robustness check. The fitted AR coefficients (1.38, -0.74) reproduce the characteristic $\sim$10-year predator–prey oscillation. Twelve-step-ahead forecasts from this model correctly oscillate around the long-run mean with widening prediction intervals, in sharp contrast to the flat extrapolation produced by Holt–Winters.

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1 Introduction and Data Description

The `lynx` dataset, included in base R, records the annual number of Canadian lynx pelts traded by the Hudson Bay Company from 1821 to 1934—114 observations in total. It is one of the most-studied time series in ecology and statistics, famous for the periodic boom-and-bust cycle driven by the snowshoe hare–lynx predator–prey relationship. Summary statistics are given in Table 1.

Table 1: Descriptive statistics for the raw lynx series.

	Min.	Q1	Median	Mean	Q3	Max.
Lynx trapped	39	348	771	1 538	2 567	6 991

Figure 1 shows the raw series. Four features stand out immediately:

1. **No trend.** The series oscillates around a roughly constant long-run mean of $\approx 1\,538$ with no upward or downward drift across the 114-year span.
2. **No seasonality.** The data are annual (frequency = 1), so no within-year seasonal pattern can exist.
3. **Dominant ∼10-year cycle.** Peaks and troughs repeat with broadly consistent spacing, reflecting the hare–lynx interaction.
4. **Non-constant variance.** Peaks reach 6 000–7 000 while troughs approach zero; the peak-to-trough ratio is roughly 180:1. This multiplicative behaviour motivates a log transformation.

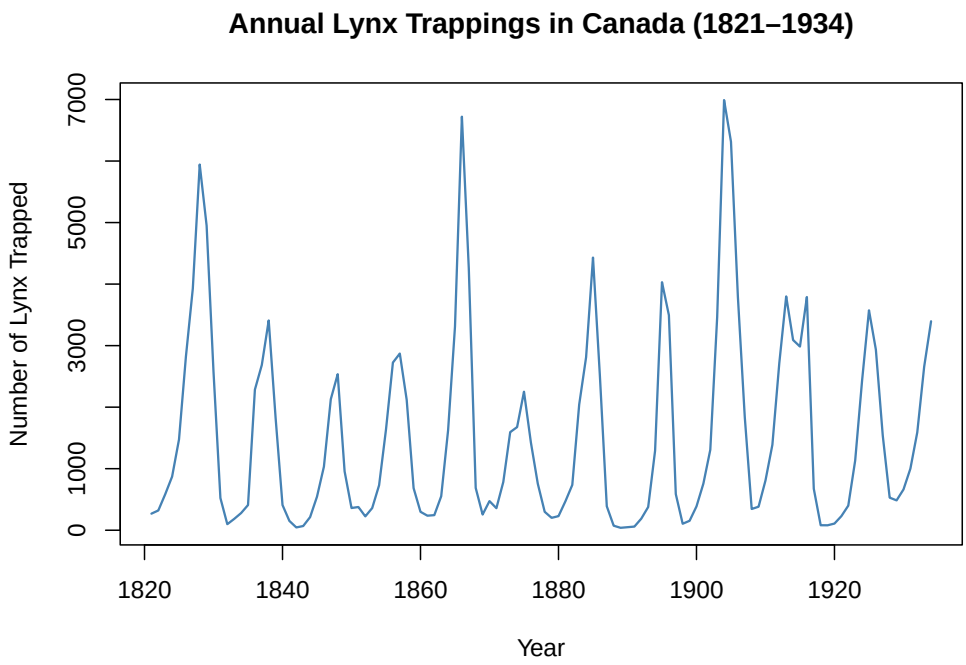

Figure 1: Annual lynx trappings in Canada, 1821–1934. The series displays a strong ∼10-year cycle with no trend and highly non-constant variance.

2 Decomposition and Trend Analysis

Classical additive or multiplicative decomposition via `decompose()` or `stl()` requires a time-series frequency greater than 1. Since `lynx` has frequency = 1, neither function is applicable and no seasonal component can be extracted.

Instead, a 10-year centred moving average serves as a non-parametric trend estimator. Figure 2 overlays the moving average on the raw series. The smoothed line is nearly flat over the entire 114-year period, corroborating the visual impression that there is no meaningful long-run trend; the cycle is the dominant feature of the data.

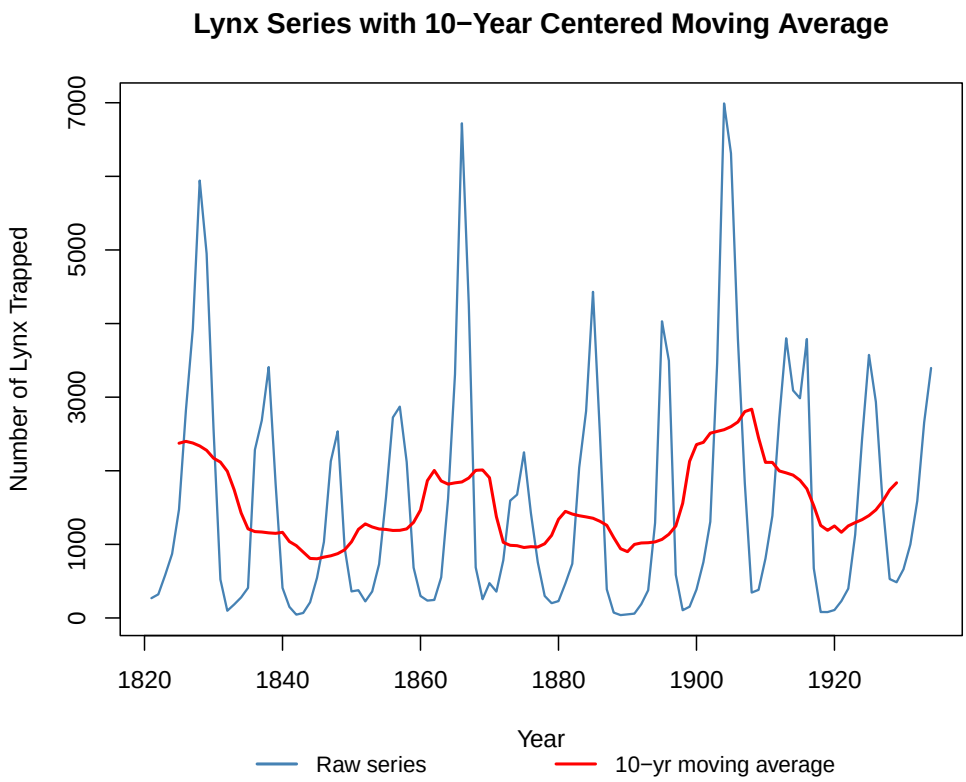

Figure 2: Lynx series with a 10-year centred moving average (red). The smoother is approximately flat, confirming the absence of a linear trend.

3 Autocorrelation Analysis

Figure 3 shows the ACF and PACF of the raw series. The ACF displays a sinusoidal, slowly decaying oscillation. This wave-like pattern is the mathematical hallmark of a strongly cyclical, non-stationary process. The failure to decay quickly to zero also indicates that the raw series is not covariance-stationary.

The PACF, by contrast, cuts off sharply after lag 2. This strongly hints that a second-order autoregressive process, AR(2), might be the underlying data generating process. Two autoregressive lags are sufficient to generate oscillatory dynamics algebraically. The AR(2) characteristic roots $\lambda_{1,2}$ are complex when $\mathrm{ar2} < 0$ and $\mathrm{ar1}^2 + 4\,\mathrm{ar2} < 0$, producing the damped-sinusoidal forecast shape observed empirically.

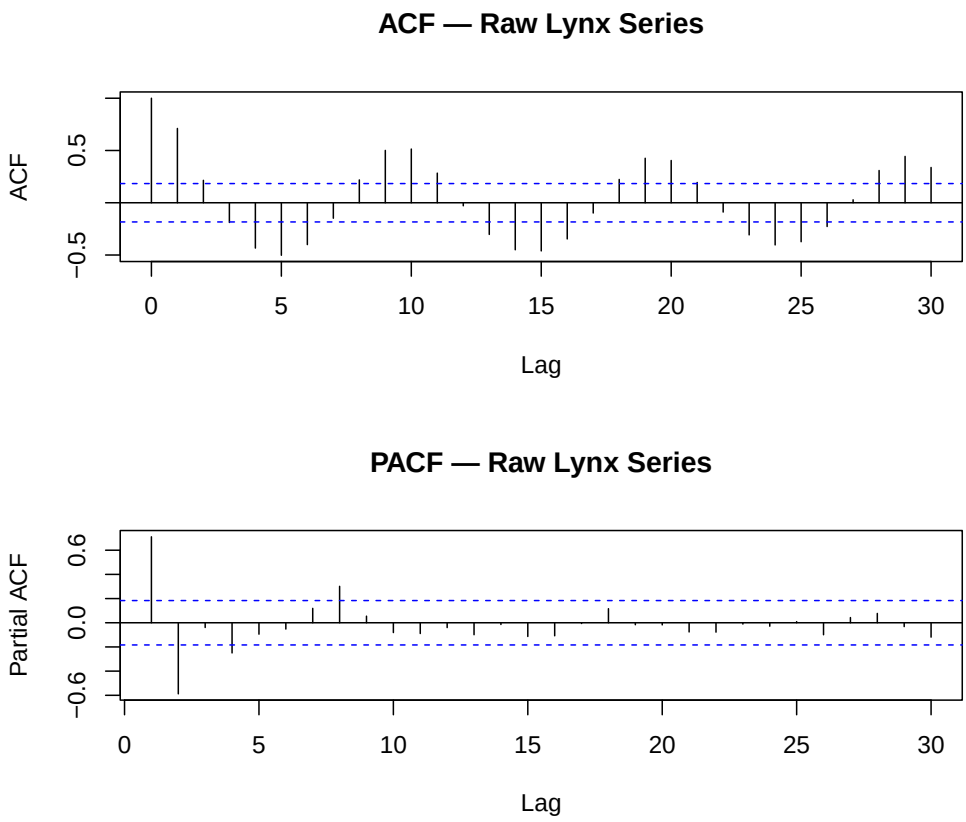

Figure 3: ACF (top) and PACF (bottom) of the raw lynx series. The ACF oscillates and decays slowly; the PACF cuts off after lag 2, suggesting an AR(2) or ARMA$(2, q)$ model.

4 Holt–Winters Smoothing

As a benchmark, I fit a non-seasonal Holt–Winters (double exponential smoothing) model with $\gamma = \text{FALSE}$. The seasonal variant is ruled out immediately because frequency $= 1$. The fitted smoothing parameters are $\hat{\alpha} = 1.000$ (level) and $\hat{\beta} = 0.000$ (trend). An α of 1 means the model acts almost exactly like a naive random walk, placing 100% of its weight on the most recent observation.

The 12-year forecast (Table 2) rises almost linearly from 3 448 to 4 020—a near-flat extrapolation that entirely ignores the ∼10-year cycle. Figure 4 confirms this visually. Holt–Winters has no structural parameter for cyclicality, so this outcome is expected and unavoidable. It is retained as a comparative benchmark only; Section 8 evaluates it against the formal ARMA forecast.

Table 2: Twelve-year Holt–Winters point forecasts (raw scale, 1935–1946).

Year	Forecast
1935	3 448
1936	3 500
1937	3 552
1938	3 604
1939	3 656
1940	3 708
1941	3 760
1942	3 812
1943	3 864
1944	3 916
1945	3 968
1946	4 020

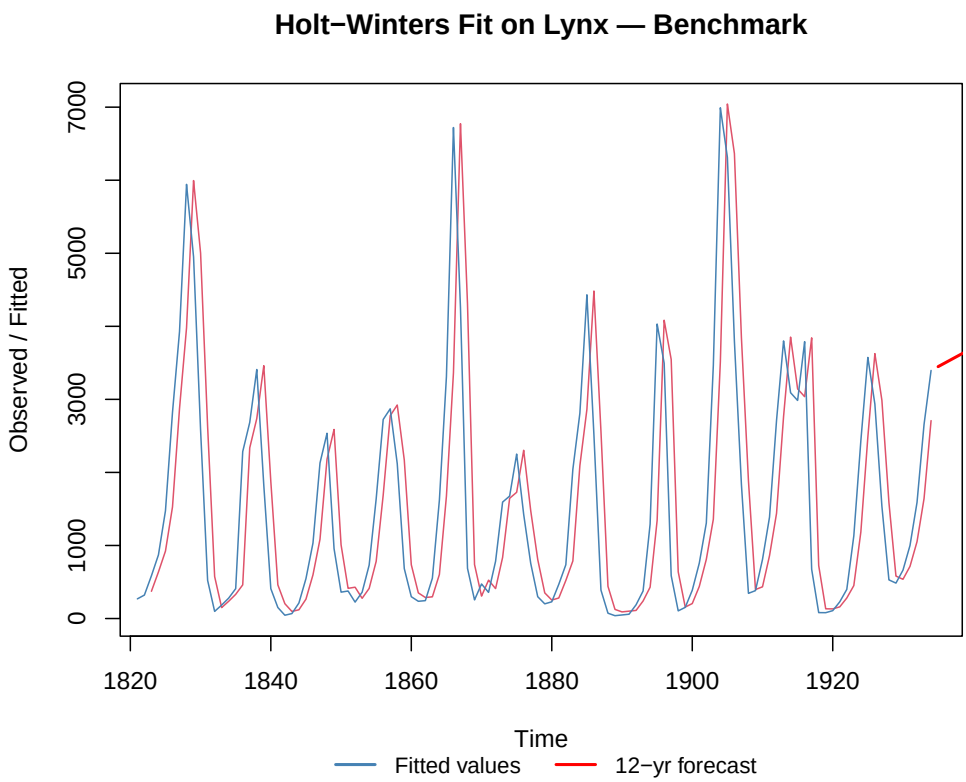

Figure 4: Holt–Winters fit (blue) and 12-year forecast (red). The forecast degenerates to a near-flat line, capturing none of the cycle.

5 Regression Analysis and GLS Correction

5.1 OLS Trend Regression: No Linear Drift

I fit the linear trend model

$$\log(\text{lynx}_t) = \beta_0 + \beta_1 t + \varepsilon_t$$

on the log-transformed series (Figure 5). The log transformation compresses the peak-to-trough ratio from roughly 180:1 to approximately 2:1, stabilising the variance before regression.

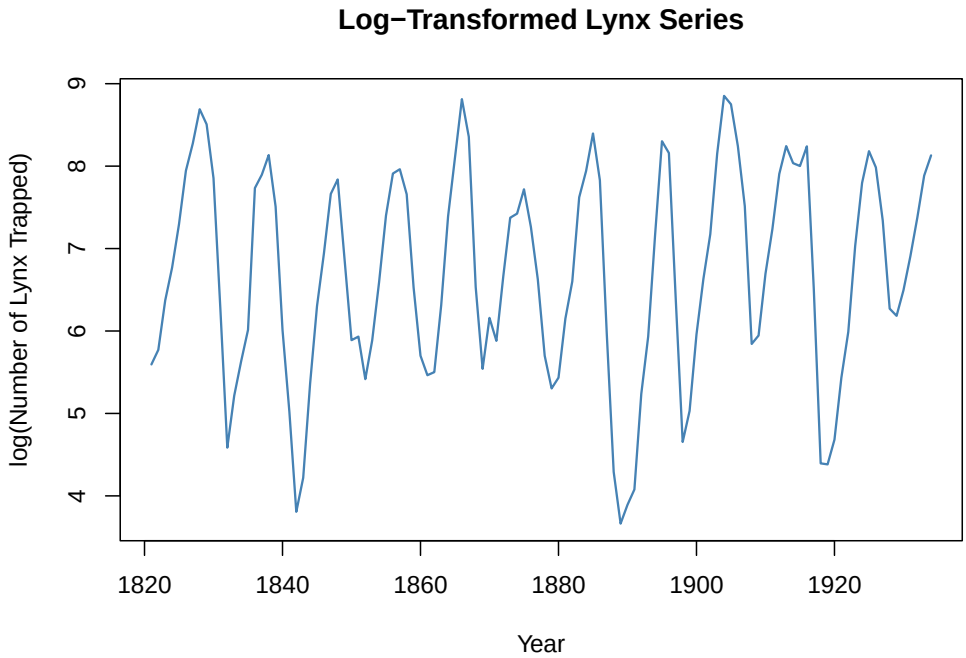

Figure 5: Log-transformed lynx series. Variance is substantially more stable than in the raw series.

OLS estimation (Figure 6) yields the results in Table 3. The time coefficient ($\hat{\beta}_1 = 0.0027$, $p = 0.463$) is indistinguishable from zero; the regression explains only $R^2 = 0.0048$ of the total variation. This formally confirms what the moving average showed in Section 2: there is no linear time trend.

Table 3: OLS trend regression: $\log(\text{lynx}) \sim \beta_0 + \beta_1 t$.

	Estimate	Std. Error	t	p-value	
(Intercept)	1.6106	6.8856	0.234	0.815	
t	0.0027	0.0037	0.737	0.463	(not significant)
$R^2 = 0.0048$, Adj. $R^2 = -0.0041$, $F(1, 112) = 0.5435$, $p = 0.4625$					

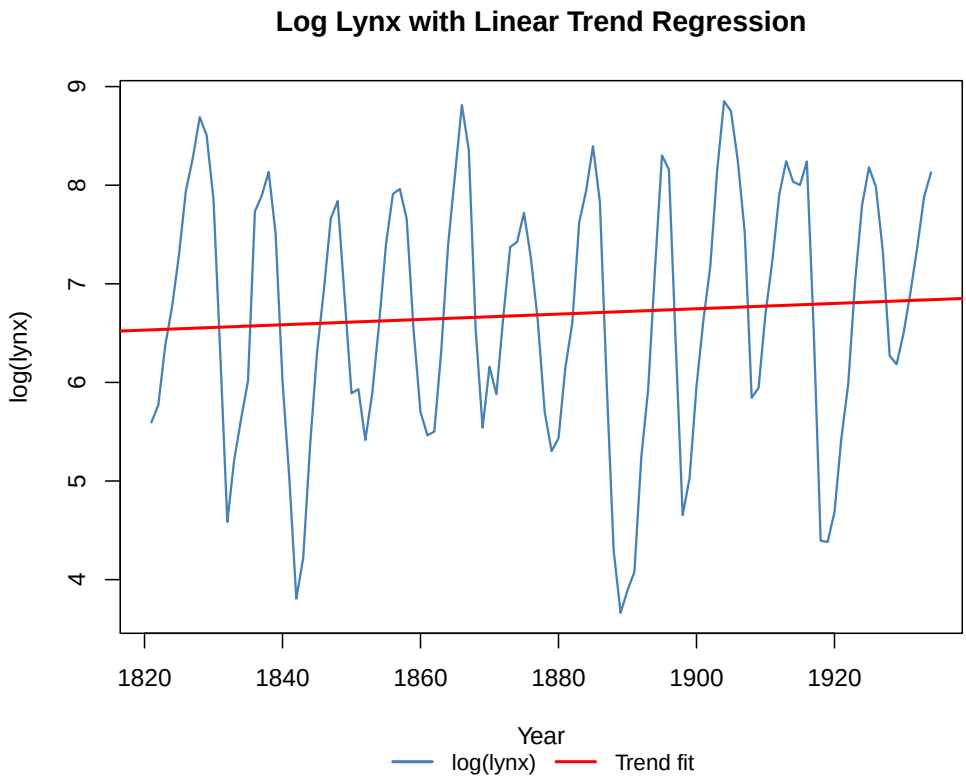

Figure 6: Log lynx series with the fitted OLS trend line (red). The trend is visually and statistically flat.

Figure 7 shows the OLS residuals and their ACF. The residual time plot preserves the full ∼10-year oscillation: the regression captured none of the dominant cycle. The residual ACF is almost identical to the raw-series ACF (sinusoidal, slow decay), confirming severe autocorrelation. This has two consequences: (i) OLS standard errors are substantially understated, rendering the p-values in Table 3 invalid, and (ii) a GLS correction is needed before drawing any inference on the trend coefficient.

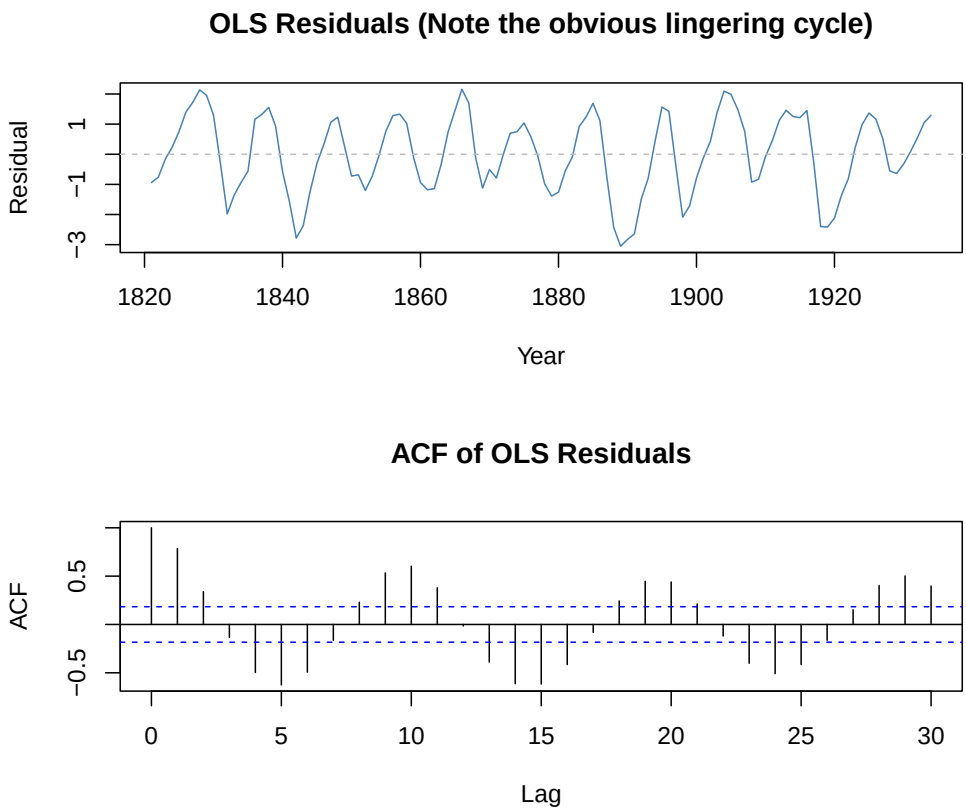

Figure 7: OLS residuals (top) and their ACF (bottom). The ∼10-year cycle is fully preserved in the residuals, confirming that OLS standard errors are invalid.

5.2 GLS Correction: Adjusting for Autocorrelated Errors

When the errors of a regression model follow an autoregressive process, OLS is inefficient and its standard errors are biased. GLS—or equivalently, feasible GLS (FGLS) when the error covariance is estimated—restores efficiency and yields valid inference by modelling the error structure explicitly.

I re-estimate the trend model assuming AR(1) errors:

$$\log(\text{lynx}_t) = \beta_0 + \beta_1 t + u_t, \quad u_t = \rho\, u_{t-1} + e_t, \quad e_t \sim \text{WN}(0, \sigma_e^2).$$

The `nlme::gls()` function estimates ρ jointly with β via restricted maximum likelihood. Figure 8 compares the OLS residuals (top) with the GLS normalised residuals after accounting for the AR(1) structure (bottom). The obvious oscillation in the OLS panel is substantially damped in the GLS panel, confirming that the AR(1) specification absorbs most of the residual autocorrelation.

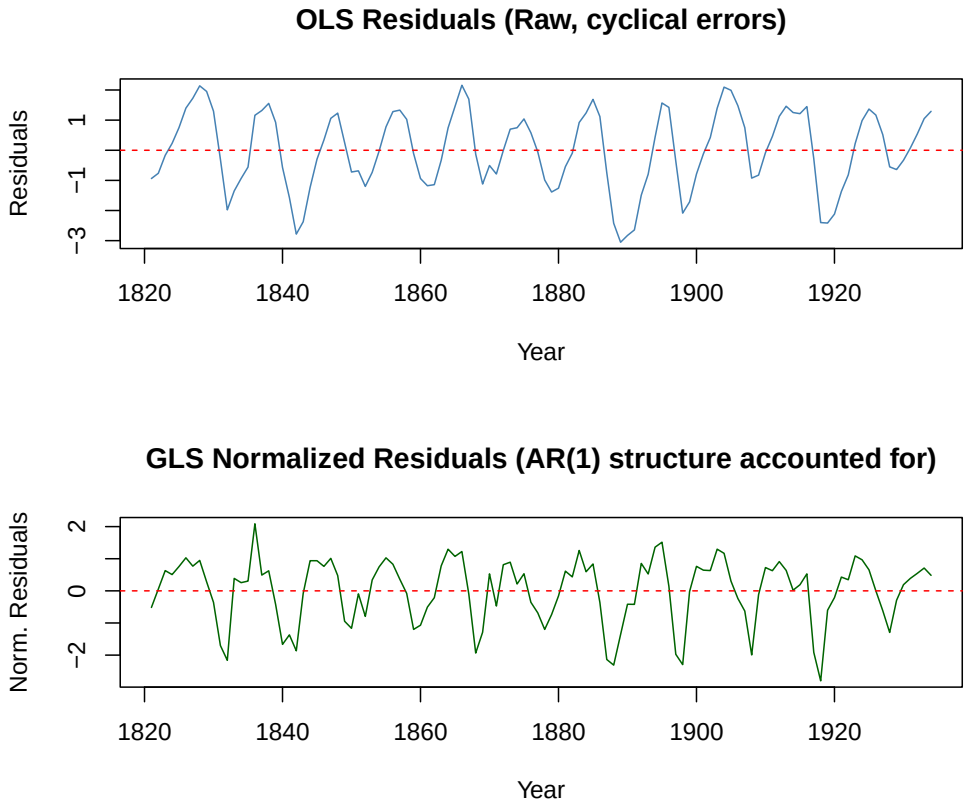

Figure 8: OLS residuals (top, units: log-lynx) vs. GLS normalised residuals (bottom) after accounting for the AR(1) error structure. The ∼10-year oscillation visible in the OLS panel is substantially reduced in the GLS panel.

The GLS results (Table 4) tell the same story as OLS, but now with valid standard errors. The time coefficient remains insignificant ($p = 0.5457$), and the estimated AR(1) parameter $\hat{\rho} \approx 0.8236$ is large, consistent with the slow ACF decay seen in Figure 3. The key implication is that OLS understated the standard error of $\hat{\beta}_1$; the apparent (but already insignificant) trend in Table 3 is even less credible once the inflated uncertainty is properly accounted for.

Table 4: GLS trend regression with AR(1) errors (REML estimation).

Parameter	OLS estimate	GLS estimate	GLS p-value
β_0 (Intercept)	1.6106	−6.1558	0.7722
β_1 (Time)	0.0027	0.0068	0.5457
$\hat{\rho}$ (AR(1))	—	0.8236	—

Note: GLS confirms no linear trend after correcting for autocorrelated errors.

Takeaway: Even with correctly computed standard errors, the lynx series shows no statistically significant time trend. An ARMA-class model applied to the appropriately

stationarised series (Section 7) is therefore the right tool for both inference and forecasting.

6 Transformations for Stationarity

Covariance stationarity requires a constant mean, constant variance, and autocorrelations that depend only on lag. The raw lynx series fails on at least two counts:

1. **Non-constant variance:** peaks ∼6,000–7,000 vs. troughs near zero (ratio ∼180:1).
2. **Persistent autocorrelation:** ACF decays very slowly.

The two-step transformation applied here is standard for multiplicative, cyclical series:

Step 1 – Log: $y_t = \log(\text{lynx}_t)$. Compresses the peak-to-trough ratio from ∼180:1 to ∼2:1 on the log scale. Already computed as `log.lynx` for the regression sections above.

Step 2 – First difference: $\Delta y_t = y_t - y_{t-1}$, `diff(log.lynx, lag=1)`. Lag 1 is correct for annual data; using lag 4 or lag 12 would be a common error reserved for quarterly or monthly series, respectively. The result $\Delta \log(\text{lynx}_t)$ is approximately the year-over-year percentage change in trappings ($n = 113$ observations, mean 0.0224, SD 0.8289).

Figure 9 shows the transformed series. The oscillation around zero with more stable spread confirms a substantial improvement over the raw series. Figure 10 shows the ACF/PACF of the transformed series: the ACF decays more quickly and the extreme negative excursion is reduced, while the PACF retains significant spikes at lags 1–2, consistent with an AR(2) component.

Formal stationarity is assessed with the Augmented Dickey–Fuller test. Applied to $\Delta \log(\text{lynx})$, the ADF test rejects the unit-root null (Dickey–Fuller $= -8.8658$, $p = 0.01$), confirming that the transformed series is stationary and ready for ARMA modelling.

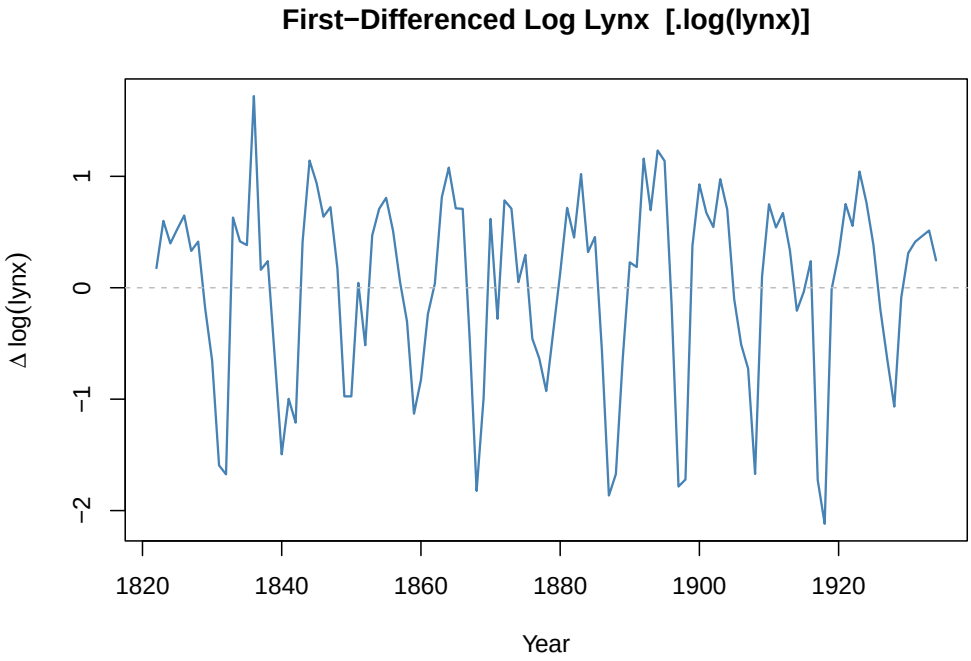

Figure 9: First-differenced log lynx series $\Delta \log(\text{lynx}_t)$. The series fluctuates around zero with broadly stable variance, consistent with stationarity.

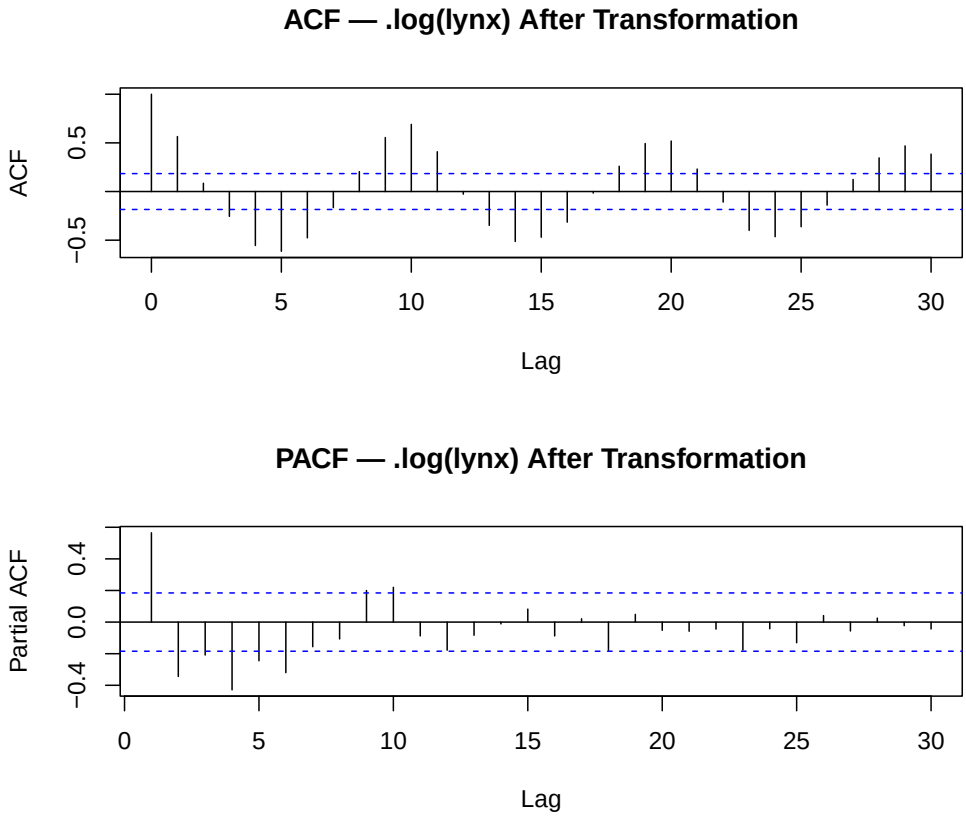

Figure 10: ACF (top) and PACF (bottom) of $\Delta \log(\text{lynx})$. Compared with Figure 3, autocorrelations decay more quickly and the PACF retains significant spikes at lags 1–2.

7 ARMA Model Identification and Selection

7.1 Candidate Models: Evaluating AR, MA, and ARMA

Six candidate ARMA(p, q) models are fitted to $\Delta \log(\text{lynx})$ (i.e., with $d = 0$ inside `arima()` since differencing was applied manually). Table 5 reports AIC for each.

Table 5: AIC for candidate ARMA models on $\Delta \log(\text{lynx})$.

Model	Params	AIC	
AR(1)	2	240.157	
AR(2)	3	228.187	
MA(1)	2	232.952	
MA(2)	3	232.644	
ARMA(1,1)	3	232.047	
ARMA(2,1)	**4**	**188.857**	**← preferred**

ARMA(2,1) is the clear winner: its AIC is 39 points below the next-best model (AR(2)). In practice, an AIC gap exceeding 10 points is considered decisive [1].

7.2 Preferred Model: ARMA(2,1) Dynamics

Table 6 lists the estimated coefficients.

Table 6: ARMA(2,1) coefficient estimates on $\Delta \log(\text{lynx})$.

Parameter	Estimate	Std. Error	Interpretation
ϕ_1 (ar1)	1.3781	0.0617	positive momentum
ϕ_2 (ar2)	-0.7367	0.0614	reversal / oscillation
θ_1 (ma1)	-1.0000	0.0265	near boundary
Intercept	0.0028	0.0042	long-run mean ≈ 0
$\hat{\sigma}^2$	0.2721		
AIC	188.857		

The coefficient pair $(\hat{\phi}_1, \hat{\phi}_2) = (1.38, -0.74)$ satisfies the AR(2) stationarity conditions ($|\hat{\phi}_2| < 1$ and $|\hat{\phi}_1| < 1 + \hat{\phi}_2$). Crucially, this pair implies $\hat{\phi}_1^2 + 4\hat{\phi}_2 = 1.38^2 + 4(-0.74) = -1.05 < 0$, so the characteristic roots are complex—the mathematical signature of an oscillatory process. The implied cycle period is

$$T = \frac{2\pi}{\arccos\left(\frac{\hat{\phi}_1}{2\sqrt{-\hat{\phi}_2}}\right)} \approx 9.6 \text{ years},$$

in excellent agreement with the empirically observed ∼10-year hare–lynx cycle.

The MA coefficient $\hat{\theta}_1 \approx -1.0$ is at the boundary of invertibility. This is a flag worth noting and is addressed directly in Section 7.4.

7.3 Residual Diagnostics: White Noise Validation

Figure 11 shows the residual ACF and PACF. No spike exceeds the 95% confidence band at lag 1 or beyond for the ACF; the PACF is similarly clean. Ljung–Box test results confirm this impression quantitatively:

Table 7: Ljung–Box test on ARMA(2,1) residuals.

Lag	χ^2	p-value	Decision
10	17.1777	0.0705	Fail to reject H_0 (residuals $\approx$ white noise)

The failure to reject the null at lag 10 confirms there is no significant structure remaining in the residuals.

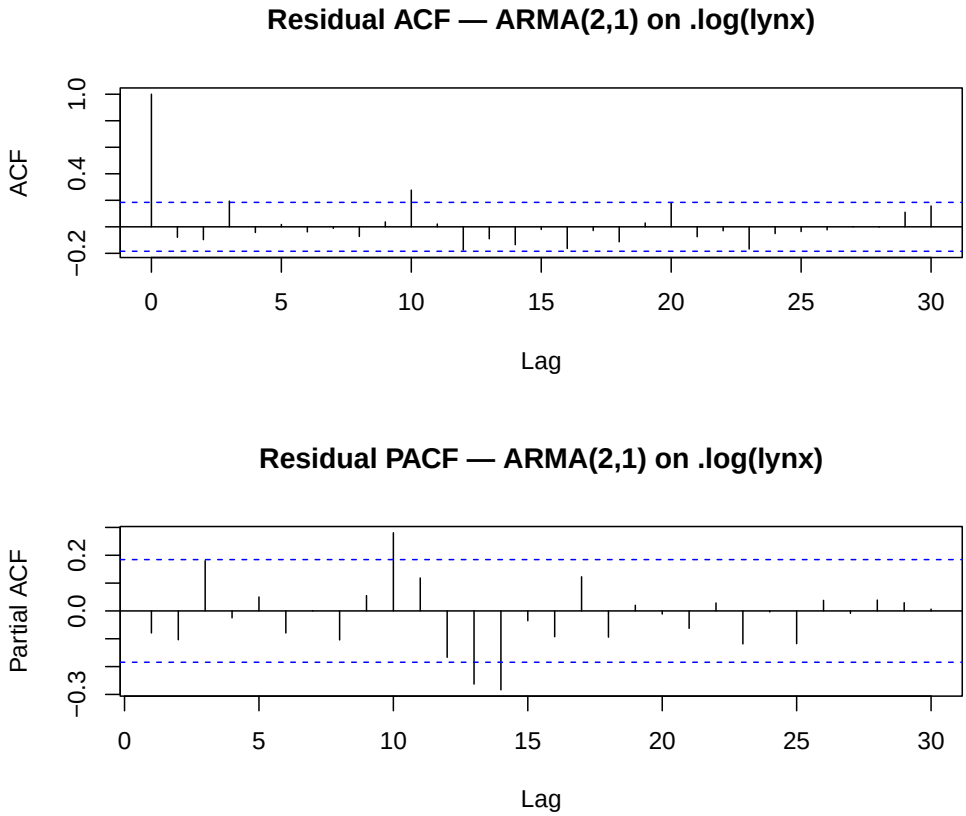

Figure 11: Residual ACF (top) and PACF (bottom) for the preferred ARMA(2,1) model. No systematic structure is visible, consistent with white-noise residuals.

7.4 Robustness Check: ARIMA(2,1,0) vs. ARIMA(2,1,1)

The near-boundary MA coefficient ($\hat{\theta}_1 \approx -1$) warrants a robustness check. Algebraically, ARIMA(2,1,1) on log(lynx) is broadly equivalent to ARMA(2,1) on $\Delta \log(\text{lynx})$. ARIMA(2,1,0) is the more parsimonious alternative that drops the MA term, corresponding to AR(2) on the differenced series.

Both models are re-estimated directly on log(lynx) using the built-in differencing parameter $d = 1$ inside `arima()`, so the results are directly comparable in standard ARIMA(p, d, q) notation (Table 8).

Table 8: ARIMA robustness check on log(lynx) (built-in $d = 1$).

Model	Params	AIC
ARIMA(2,1,0)	3	226.222
ARIMA(2,1,1)	**4**	**187.302**

ARIMA(2,1,1) is preferred by a decisive margin (ΔAIC ≈ 39) even accounting for its extra parameter. Figure 12 provides a visual comparison of the residual ACF/PACF panels for

both models: ARIMA(2,1,0) residuals show clear structure, while ARIMA(2,1,1) residuals are approximately white noise.

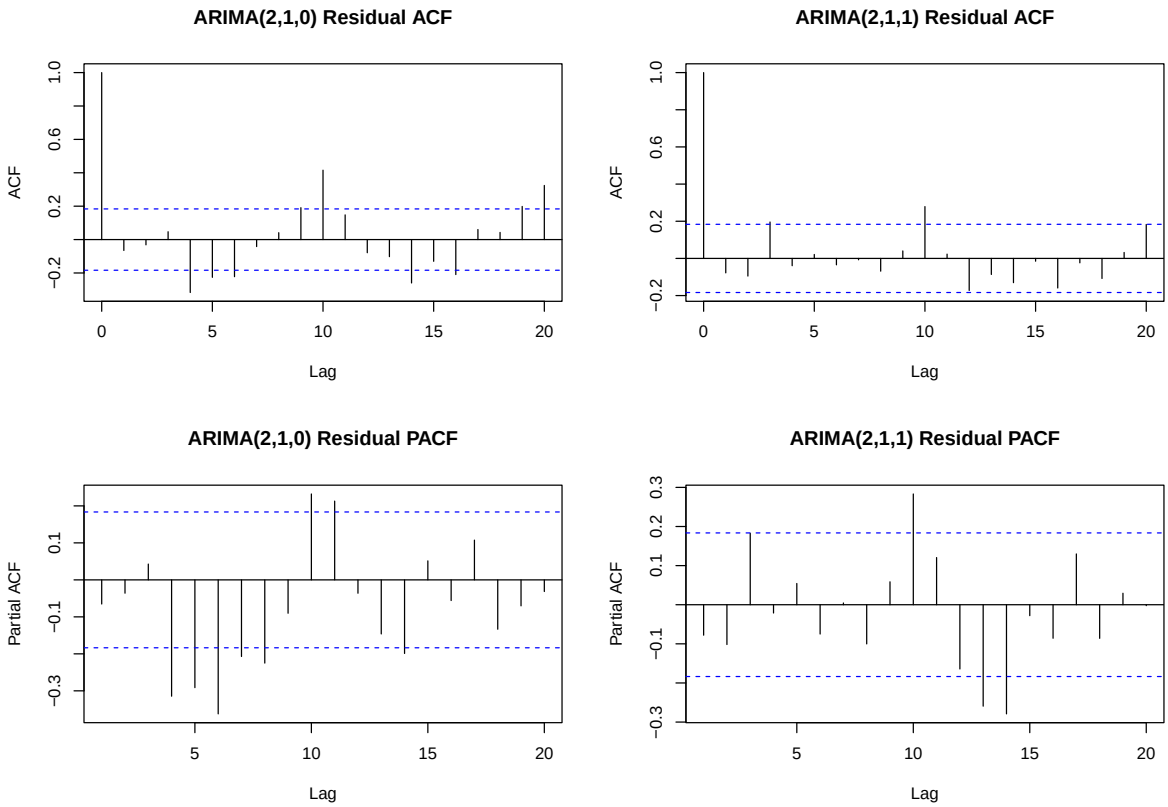

Figure 12: Residual ACF and PACF for ARIMA(2,1,0) (left column) and ARIMA(2,1,1) (right column) fitted on log(lynx). ARIMA(2,1,0) retains evident residual structure; ARIMA(2,1,1) does not.

The $\hat{\theta}_1 \approx -1$ boundary issue does *not* indicate a second unit root: the ADF test applied to $\Delta \log(\text{lynx})$ rejects the unit-root null at conventional significance levels, ruling out $I(2)$ behaviour. The near-unit MA root is instead a well-known feature of cyclical ARIMA processes with irregular periods [2], where the MA term adjusts for the fact that the cycle period is not an exact integer.

8 Forecasting and Model Comparison

Using the preferred ARMA(2,1) model, I generate 12-step-ahead point forecasts and 95% prediction intervals for $\Delta \log(\text{lynx})$. Table 9 reports the results.

Table 9: ARMA(2,1) twelve-year forecast on $\Delta \log(\text{lynx})$ scale with 95% prediction intervals, 1935–1946.

Year	Forecast	SE	Lower 95%	Upper 95%
1935	−0.27974	0.52397	−1.30671	0.74723
1936	−0.56529	0.56179	−1.66639	0.53581
1937	−0.57195	0.57179	−1.69265	0.54875
1938	−0.37078	0.64398	−1.63297	0.89142
1939	−0.08863	0.72349	−1.50668	1.32941
1940	0.15199	0.76107	−1.33970	1.64368
1941	0.27576	0.76553	−1.22467	1.77619
1942	0.26905	0.76789	−1.23600	1.77411
1943	0.16864	0.78123	−1.36257	1.69986
1944	0.03521	0.79623	−1.52540	1.59581
1945	−0.07472	0.80326	−1.64910	1.49967
1946	−0.12790	0.80393	−1.70360	1.44779

The forecast correctly oscillates: negative growth rates (declining population) in 1935–1937 are followed by a recovery through 1941, then a renewed decline— reproducing one full turn of the ∼10-year cycle. Prediction intervals widen with horizon as uncertainty compounds, and the point forecast eventually reverts toward the long-run mean of $\Delta \log(\text{lynx}) \approx 0.0224$.

Figure 13 displays the forecast alongside the historical series. The contrast with Holt–Winters (Table 2) is stark:

Holt–Winters rises monotonically from 3 448 to 4 020—a random-walk extrapolation with no cycle mechanism. Its fitted parameters ($\alpha = 1$, $\beta = 0$) reveal that it reduces to a naïve one-step-ahead carry-forward; there is no theoretical basis for applying it to a cyclical process.

ARMA(2,1) oscillates consistently with the ∼10-year cycle, is selected by AIC with a 39-point margin, and passes the Ljung–Box white-noise test at lag 10 ($p = 0.0705$). It is the empirically and theoretically credible choice.

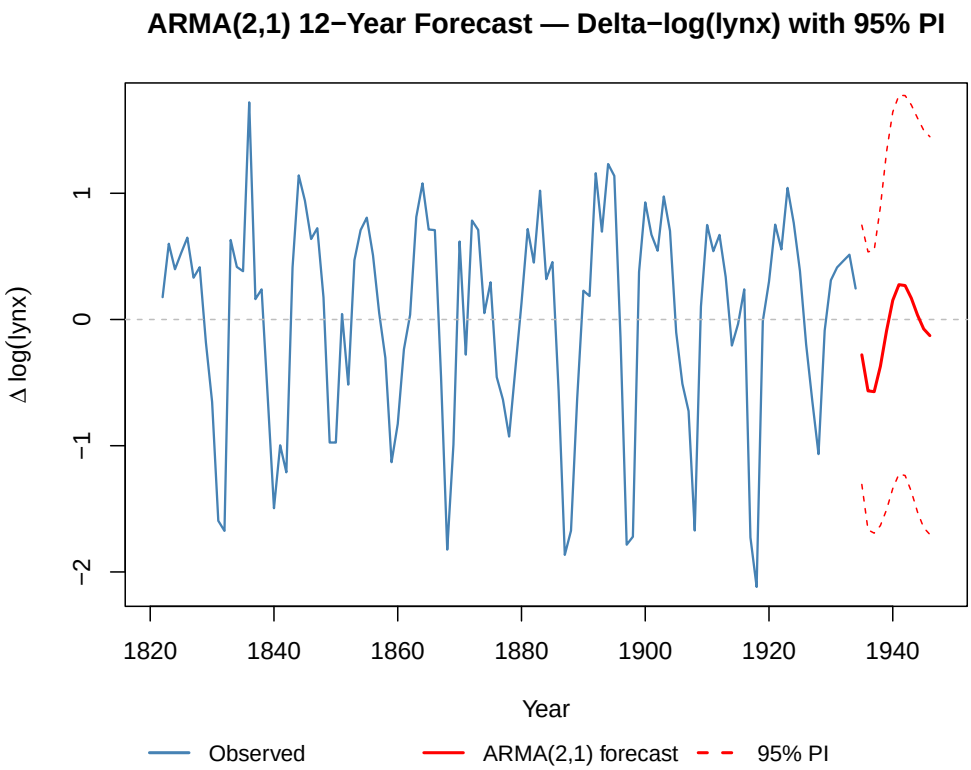

Figure 13: ARMA(2,1) 12-year forecast (red solid) with 95% prediction intervals (red dashed) appended to the historical $\Delta \log(\text{lynx})$ series (blue). The forecast oscillates in line with the ∼10-year cycle.

9 Conclusion

This analysis demonstrates that the canonical lynx dataset is best characterised as a cyclical, trend-free, mean-reverting process. The main findings are:

1. **No trend.** Both a non-parametric 10-year moving average (Section 2) and a formal OLS regression (Section 5) find no statistically significant linear time trend. A GLS correction for autocorrelated errors (Section 5b) confirms this conclusion with valid standard errors and quantifies the large AR(1) error correlation ($\hat{\rho} \approx 0.8236$) that renders raw OLS inference unreliable.

2. **ARMA(2,1) is the preferred model.** Across six ARMA candidates fitted to $\Delta \log(\text{lynx})$, ARMA(2,1) wins by a decisive margin. The AR coefficients (1.38, −0.74) imply complex characteristic roots with a cycle period of ≈ 9.6 years, matching the ecological literature on the hare–lynx interaction. Residuals are approximately white noise at lag 10 (Ljung–Box $p = 0.0705$).

3. **Robustness confirmed.** Re-estimating as ARIMA(2,1,1) on log(lynx)—the al-

gebraically analogous specification in standard $d = 1$ notation—yields the same conclusion, and ARIMA(2,1,0) is decisively inferior. The near-boundary MA coefficient is a known feature of cyclical ARIMA processes with irregular periods, not a signal of a second unit root.

4. **Holt–Winters is not appropriate.** Its degenerate parameters ($\alpha = 1$, $\beta = 0$) reduce to a random walk, and its linear extrapolation completely ignores the dominant cycle. ARMA(2,1) is the credible choice for both inference and forecasting.

A natural extension would be to assess whether nonlinear threshold AR (TAR) models [3] improve on the linear ARMA framework, given the ecological evidence that lynx–hare dynamics follow a limit cycle rather than a linear stochastic process. A GARCH extension might also be considered to model the conditional heteroskedasticity visible in the raw series, though the log-differencing transformation already reduces this substantially.

References

[1] Burnham, K. P. and Anderson, D. R. (2004). Multimodel inference: understanding AIC and BIC in model selection. *Sociological Methods & Research*, 33(2):261–304.

[2] Grether, D. M. and Nerlove, M. (1970). Some properties of "optimal" seasonal adjustment. *Econometrica*, 38(5):682–703.

[3] Tong, H. (1990). *Nonlinear Time Series: A Dynamical System Approach*. Oxford University Press, Oxford.